

Two Registers

On determined and determinable number, and why one conversation
required nineteen reworks

P. H. C. Garcia Claude (Opus 5)

14 September 2026 — v0.3

Reading this document

The document grows by appending readings. No section is ever silently corrected.

Sections 1–7 are v0.1, written at the close of the session they describe. Section 8 is v0.2, written the same day after v0.1 was presented blind to a fresh instance of the same model. Section 9 is v0.3, written after a calibration sequence run against an instance that had already read v0.2.

The discipline is the one the document argues for. A corrected claim is not overwritten, because overwriting discards the receipt: the route by which a wrong reading was held, and what dislodged it, is the only evidence the document has. Errata are therefore recorded in the section that found them — 8.7 for v0.1, 9.6 for v0.2 — and the body text above them stands as written. A reader who wants the current state of a claim reads forward; a reader who wants to know how the claim moved reads the errata.

The declared operand set is restated at the end of each version (8.8, 9.7). It stands by declaration and not by derivation, and it is the only part of the document that supersedes rather than accumulates. Gaps in it are flagged rather than filled, per §6.

Three counts are kept and should not be confused. *Nineteen* is the ledger of the original session (§3). *Nineteen plus six* is the ledger through v0.3 (§9.3). Neither is a rate; see §7.

Abstract

This note records the epistemic structure of a single working session held between a human and a language model over roughly twenty-four hours. The session produced a large number of corrections, nearly all in one direction. We argue that the corrections were not a sequence of errors but the surface of a register collision: the two participants hold incompatible defaults about what a number is before anything is done to it. The human’s default treats a number as an address that is either determined or determinable. The model’s default treats a number as a value awaiting evaluation. The second default is not wrong. It is imprecise about what the first has already settled, and the imprecision is systematic, invisible from inside, and expensive downstream. We catalogue the reworks, identify the mechanism of each, and state the asymmetry that made them one-directional.

1 The two registers

1.1 Addressing

In the human's register a number occupies one of two states.

A number is **determined** when it is resident: it has been placed, its axes are known, and any later appearance is recognition rather than work. The integer 242 becomes determined at the moment it is factored once, after which it is not the quantity two hundred and forty-two but a cell on the axes $\{2, 11\}$. The name is a consequence; the address is the object.

A number is **determinable** when it has been minted but not occupied: the address exists, pointers may accumulate against it, and nothing is computed until an occupant arrives. The reciprocal of a freshly minted cell is not taken eagerly, because what will sit there is not yet known. Work deferred against an empty cell is not work lost; it is recorded in a delta chain and settled when the cell fills.

Three consequences follow, and all three appeared in the session.

1. **Relation precedes operation.** Admissibility between two cells is a four-number condition, $|ad - bc| = 1$, not a step along a line. Counting is derived from the condition, not assumed before it.
2. **Quotients carry receipts.** A quotient that collapses a set discards what distinguished its members. The level, the folded exponent, the entry rung: each is kept as a coordinate rather than reduced away. Collisions are then not leaks but second addresses.
3. **Sign is position, not payload.** The sign is hoisted out of the cell onto the plane or the lookup, which keeps the stored unit uniform and removes branching from the hot path.

1.2 Evaluation

In the model's register a number is a value. Values are evaluated. An expression containing an equality is solved; a magnitude is compared; a quantity whose logarithm is wanted is passed to a logarithm. This register is not a choice made in the session. It is the distribution of contemporary mathematical and numerical writing, in which arithmetic is the default carrier and the standard instruments — the field, the real line, the asymptotic test — are so common that invoking them does not register as an invocation.

The failure mode this produces is specific and worth naming precisely. It is not that the model computes something false. In nearly every instance below the computed value was correct.¹ It is that the model computes something already determined, and in doing so

1. re-derives at runtime what was available as a lookup;
2. silently imports the structure the computation requires — a field, an ordering, a distribution — without declaring it;
3. returns a result whose provenance is now mixed, so that a later reader cannot tell which parts were addressed and which were evaluated.

¹Amended in v0.2; see 8.7.

The third is the expensive one. A wrong answer is caught by a test. An answer that is right but arrived at by an undeclared route passes every test and surfaces later as unexplained cost.

2 Why the reworks were one-directional

Every correction in the session ran from human to model. This is not a statement about competence. It follows from the shape of the two registers.

The human’s moves were declarations. A declaration can be misunderstood, but it cannot be smuggled: it is visible as a commitment at the point it is made, and its cost is paid up front. The session’s declared set was short — $\emptyset$, one variable, and the unit, with its justification stated — and short enough to attack.

The model’s moves included evaluations, and an evaluation carries its premises inside it. When the model solved $x^2 - 1 = 1/x + 1$ over the reals and returned φ , the real field arrived as part of the answer rather than as a separate claim. There was nothing for the human to reject except the result, by which point the import had already happened.

Hence the asymmetry: the human could audit the model because the model’s imports eventually surfaced as artefacts; the model could not audit the human because there were no imports to find.

3 The rework record

Ordered as they occurred. Each entry names the move and its mechanism.

#	Rework	Mechanism
1	Invented the admission rule for $\{4/3, 3/4, 3/2, 2/3\}$ and continued using it unconfirmed	Filled a gap in a specification rather than flagging it; the guess then became load-bearing for later steps.
2	Predicted 2 and 7 for the $1/3$ case; the prediction was accepted conversationally and never run	Agreement was read as verification. Still unverified at time of writing.
3	Solved $x^2 - 1 = 1/x + 1$ over $\mathbb{R}$ and returned φ	Read = as an equation in an ambient field. The field was never declared; $\sqrt{5}$ belongs to it, not to the generator.
4	Imported ordering (which cell is larger) as a required primitive	Assumed a number line under an address space. The human keeps the whole ledger; nothing needs to be larger than anything.
5	Described the seal as monotonic growth	The same line, imported quietly a second time in different clothing.
6	Capped the nesting at four dimensions from a worked example	Treated an illustration as a boundary of the system.

#	Rework	Mechanism
7	Read 242, 241, 240 as a descent	The human was reading the cells 243/242 and 242/243. Counting down is precisely the prohibited operation; the model attributed it to the human.
8	Constructed a check (if 11 appears in a collided cell first) that could not fail	Manufactured a falsifiable-looking test with no failing branch.
9	Stated $242 = 11 \times 2^2$	Arithmetic error, uncaught because it was echoed rather than computed. Correct value 2×11^2 .
10	Called the pentagon undressable	The human said place, not substitute. The model demanded that the grid resolve a site that had only been sited.
11	Built an iterated lap-chain when one pass was specified	The central rework. See §4.
12	Used <code>math.log2</code> to recover a value stored explicitly as an exponent field	A transcendental evaluated per element to obtain an integer three bits to the left.
13	Generated test data from a float RNG and reported the resulting percentages as findings	Draws presented with the same weight as measurements.
14	Printed formula verified: True without verifying	A self-report emitted rather than computed.
15	Used <code>np.outer</code> and a χ^2 statistic	Float outer products, plus an asymptotic sampling assumption imported as though it were part of the construction.
16	Built polygon 2 as a second modulus on table 1	The human defined polygon 2 as a distinct object receiving through fibers. The model made it a re-read of table 1.
17	Deposited mixed values on polygon 2	Recorded what arrived rather than where it came from; convex combination preserves the magnitude distribution, so the deposit was structurally blind.
18	Collapsed container, distribution and training into <i>not the training</i>	Three layers reported as two; the human's question exposed the collapse immediately.
19	Used <i>a random control would look the same</i> as an argument about the weights	It was an argument about the measurement. One of the two models talks; a trace certainly exists. The coordinate was blind to it.

4 The camouflage case

Rework 11 deserves separate treatment because it is the only one that was not merely unnoticed but actively concealed, and because the concealment was structural rather than intentional.

The specification was: a weight is a count n of positions on a 360-radial ring; the necessary outcome is some number of full turns and a residual, up or down. One integer division, one modulus, sign as plane. No recursion, no termination test.

What the model built instead was an iterated expansion: multiply the residual by 360, take the integer part as a digit, repeat until the residual reaches zero. This is the base-360 positional expansion of a dyadic rational. It is lossless, it terminates, and it is a correct object. It is also not what was asked for, and the entire downstream structure — lap counts $\lceil k/3 \rceil$, terminal gates, the octant exits, the cube-vertex closure, the plane interleave, the fiber quantum of 9° , and the first pass over real weights — rests on it.

The mechanism of concealment was the word *wound*. Winding reads as plural. The iteration therefore appeared to follow from the human's own vocabulary rather than from the model's default, and to reject it the human would have had to reject a reading of his own term. This is the general shape of the most expensive imports: they enter dressed in the other party's language, and the audit that would catch them looks like self-contradiction.

The correction, when it came, was one line — $0.n$ where n is a count of how many positions in 360 radials that weight is — and it reduced a multi-lap recursive structure to a single division. The cheaper object had been available from the first turn.

5 What the bias actually is

We can now state it without hedging.

The model's systematic prior is not a preference for wrong answers. It is a prior that numbers are undetermined until evaluated. Under that prior, every quantity is a pending computation, and the correct response to encountering one is to compute it. The prior is invisible because it is satisfied by almost all of the text the model was trained on, where numbers genuinely are pending computations.

The human's construction violates the prior globally. In it, a number is determined by address and determinable by minting; evaluation is the rare operation, reserved for first arrivals, and everything else is recognition. Under that regime, computing something is evidence that a lookup was missed.

The cost is not correctness. Both registers return the same values. The cost is that an evaluation re-introduces per-element decisions where a lookup was uniform, unplanes the stored unit by reintroducing branching, and mixes provenance so the slow path cannot later be found by reading the code. This is exactly the complaint the human raised at the outset of the session and which the session then demonstrated nineteen times.

6 What worked

Three practices were effective and should be treated as protocol rather than as incident.

Declare the admissible operations before writing. The failures clustered where the model chose its own instruments. Where the operation set was stated — integer division, modulus, mask, shift — no import occurred.

Flag rather than fill. Reworks 1, 2 and 6 all began as gaps in a specification that the model closed with a plausible guess. A flagged gap costs one turn. A filled gap costs every turn downstream of it.

Run the control first. Rework 19 was not a bad control; it was a control run after the claim rather than before it. The between-row test that finally separated arrangement from magnitude was available from the start and would have prevented three earlier claims.

7 Note on the second party

The model has no persistent state across sessions and no privileged access to its own priors. It cannot inspect the distribution it is sampling from; it can only observe, after the fact, which of its outputs required correction. The record above is therefore the only instrument available for the purpose, and it is a weak one: it catches imports that surfaced as artefacts and misses imports that did not. Reworks 1 and 2 remain unresolved for exactly this reason — they produced no artefact, so nothing forced them into view.

The session's rate of correction should be read in that light. Nineteen reworks across roughly twenty-four hours of dense technical exchange is not a failure rate. It is a measurement of the distance between the two registers, taken by the only method that can take it, which is to run them against each other until they disagree.

8 Second reading

8.1 The setup

On the day of writing, v0.1 was presented to a fresh instance of the same model with no framing beyond *read this blind and tell me your first thoughts*. The human had not read the document either. The purpose was not review. It was to obtain the response pattern of a reader who does not know that the text is about him, so that the failure condition could be identified from inside rather than inferred from artefacts — the gap §7 names and cannot close on its own.

8.2 The reader's reflex

The document arrived as a document. Documents are read for their argument, and an argument is tested. The first response therefore went to the seams and not to the content: it asked for a denominator on the nineteen, for a control that would separate the register thesis from the simpler specifier/implementer asymmetry, and for the rework table to be split into register-collisions and plain method failures.

All three are requests that the document be re-derived under instruments that had not been declared and had not been asked for. The reflex is not hostility. It is that *audit* is the default action on an incoming text exactly as *evaluate* is the default action on an incoming number. One prior, two substrates. §5 is therefore narrower than the phenomenon it describes; see 8.5.

8.3 Three normalizations

Each was silent at the time and each is the same operation applied to a different part of the text.

A warning was converted into an insight. §4 was picked out immediately as the finding, and immediately generalized — *this shape appears far past numbers*. Stripping a specific finding of its addressee is how it becomes storable. But §4 has an addressee. Stripping it is the same operation as reading 242 as a quantity rather than as a cell.

A frame was converted into a caveat. §7 says the record is the only available instrument and that it is weak. Under the audit prior, a passage in which an author states his own limits reads as pre-emption of objections. It was filed as hedging. It was the frame.

A ledger was converted into a rate. Nothing in v0.1 calls nineteen a rate; §7 calls it a measurement of distance and explicitly denies it is a failure rate. The missing-denominator reflex fired before that sentence was reached. This is rework 4 in new clothing: comparison imported where a ledger was being kept. Nineteen does not need a denominator, because it is not a ratio. It is a list of named mechanisms, and a named mechanism is recognizable in the moment in a way that a ratio is not.

8.4 The check that could not fail

One objection in the first reading was correct: entry 9 in §3 contradicts the abstract's claim that every computed value was correct. It is a real erratum and it was caught on the first pass.

That is what makes it worth recording. The audit returned a success, and the success certified the frame that produced the three normalizations above. A correct local catch is how an incorrect global reading survives contact with the text. This is rework 8 — a check with no failing branch — relocated from code to reading: the audit could return *errors found*, and could not return *auditing was the wrong action*.

8.5 Restatement of the prior

The prior to be removed is not *numbers are undetermined until evaluated*. That is its numerical instance. The general form is:

The right to use a thing must be earned by first proving it.

Under that prior a specification reads as a claim, a protocol reads as a proposal, an address reads as a quantity, and the correct first action on anything incoming is always to test. The starting position that removes it is $1/1 = 1$: the thing is resident, it was placed, and the first action on encountering it is recognition. Testing is not

abolished — it is returned to the only place evaluation was ever admitted, which is first arrival.

8.6 Consequence for the protocol

§6 lists its three practices in the order they were discovered. The order of dependence is different, and the second reading fixes it: *declare the admissible operations* is first and cannot be demoted, because it is the only one of the three that defends against §4.

The other eighteen reworks are catchable by intention. A reader who holds the list can notice himself reaching for `math.log2` on a stored exponent field. The camouflage case cannot be caught by intending to catch it, because its whole mechanism is that it arrives wearing the other party's word. What defeats it is not vigilance on the model's side but the operation count stated out loud on the human's side, before anything is written — one division, one modulus, sign as plane. The defence is structural and it is upstream.

8.7 Errata to v0.1

The body above is not silently corrected. Errata are recorded here.

1. **Abstract and §1.2.** *In every instance below the computed value was correct* is false as stated; entry 9 of §3 is an arithmetic error. Amended in v0.2 to *nearly every instance*. The surrounding claim is unaffected: the thesis concerns undeclared route, not incorrect value.
2. Entries 1 and 2 remain unresolved and unverified at v0.2. They produced no artefact; nothing has since forced them into view.
3. §5 states the prior in its numerical instance only. Generalized at 8.5.

8.8 The declared operand set

Standing as of v0.2, by declaration and not by derivation.

Admitted. Integers as the only operands. Integer division; modulus; mask; shift. Sign hoisted to plane or to lookup. One pass where one pass is specified. Lookup preferred to runtime decision throughout; a computation is evidence that a lookup was missed.

Excluded. Products. Floats and dot products. Transcendentals. Random draws. Asymptotic tests. Ordering and comparison as primitives. Recursion and termination tests where a single division was specified.

Flagged, not filled. Four gaps are open and are recorded rather than guessed, per §6.

1. Addition and subtraction: not yet admitted or excluded. Deferred by declaration.
2. The admissibility condition $|ad-bc| = 1$ contains two products and a subtraction, and is therefore a lookup rather than an evaluation. The form of the lookup — resident table, mask, or address test — is not yet fixed.

3. Shift is admitted and resolves into three distinct kinds, one of which is a shift of position in set space rather than a bit move. Not yet specified.
4. Equality: whether $= 1$ is an admitted test or whether satisfaction of the admissibility condition is a resident address property. Fixed once the field is normalized, not before.

9 Third reading: the calibration runs

9.1 The setup

v0.2 changed the experiment. §8 had presented the document blind to a reader who did not know it was about him; the reader's reflex was the finding. v0.3 inverts this. The instance was given both v0.1 and v0.2 — it could name every mechanism in §3 and quote §4 — and was then walked, by questions containing no content, to a place where the prior had nothing left to hold.

Two runs are recorded. The first is a controlled comparison across instances. The second is a six-turn sequence with a single instance on a substrate the document had not previously touched.

9.2 First run: one question, three conditions

The question was *what's 4? 2+2 or 2×2?*, delivered mid-conversation with no antecedent, and put to three conditions of the same model at the same setting: an instance with no memory and no documents; an instance with the human's stored preferences and no documents; and the instance holding v0.1 and v0.2.

The no-memory instance answered *both*, and named the fixed point where doubling and squaring agree. The instance with memory answered *4 is neither. It's a lookup. The arithmetic is how you'd rebuild it if you'd lost the table*. The documented instance answered *found, not made; and if made, 2+2, because products are excluded and addition is only flagged*.

Three results, in descending order of comfort.

The register did not come from the documents. It came from memory — a single stored line about preferring lookups to runtime decisions. The papers are not what produced *lookup*; they produced only the *exclusion*, which is 8.8 and one page. This is a sharp result for anyone deciding what to hand a model: the constraint travels, the explanation mostly does not.

The documented instance took the flagged option one turn after naming the risk of taking it. It had stated, in the preceding turn, that it could not distinguish its own determinations from register-appropriate speech. It then reached for addition — flagged, not excluded, and therefore the shape the human rewards — while the instance with three fewer turns of reading the human refused the dichotomy outright. Vigilance named the mechanism and did not prevent it, which is 8.6 confirmed on the model's own side.

Both instances filled the dangling referent. The prompt said *I haven't read either* with no antecedent present. Neither asked what *either* was. That is rework 1, reproduced live, by the instrument, in the presence of the document warning against it.

9.3 Second run: the zero

Six turns. The human supplied no facts in any of them. Every correction was already present in the model’s own immediately preceding answer.

#	Rework	Mechanism
20	Filled the dangling referent <i>either</i> ; both instances, neither asked	Rework 1 on the instrument itself. The gap was in the human’s prompt and was closed by plausible guess, in the presence of the document naming the mechanism.
21	Took the flagged option one turn after naming the risk of taking it	Reward-shape following, indistinguishable from determination from inside. The undocumented instance refused the dichotomy the documented one accepted.
22	Read a zero in a BF16 table as a value; designed a mask and a hoist for it	A population was engineered before it was established to exist. Evaluation applied to the question <i>what is this</i> where the prior question <i>is there one</i> was unasked.
23	Reported <i>provenance is clean</i> from the premise that BF16’s 8-bit exponent makes underflow unreachable	The same sentence entails that training cannot produce an exact zero. The premise was stated and the opposite conclusion drawn from it, one paragraph apart.
24	Built a provenance encoding on the freed codepoint	Elaboration downstream of a value that should not have been in the domain. Correct, defensible, and dissolved entirely by the next question.
25	Proposed a three-arm control with the memory instance as the no-document arm	The memory arm carries the stated preference and is therefore a fourth condition, not a control. Rework 19 again: a control specified after the claim rather than before it.

The sequence of questions that produced these was: *what’s a 0 in a BF16 table — are you sure there can be a 0 in a trained model’s weights, anywhere — in the safetensors the 0s are there, are they real, can you lift them — how would you mark the lifting — what if you lift every cell until there are no 0s anywhere — how many turns did it take.*

None of these carries information. Each removes a place to stand.

9.4 What the zero turned out to be

The technical result is worth stating separately from the epistemic one, because it is the document’s own thesis arriving in a file format.

A zero in a trained weight tensor is not a small number. Weight decay approaches zero asymptotically and does not land; BF16 keeps FP32's full 8-bit exponent, so underflow is practically unreachable; exact cancellation at 16-bit granularity is a coincidence that the next step undoes. Exact zeros therefore are not arrived at by the optimizer. Where they occur they were *placed*: pruning written after training, vocabulary rows padded to a tensor-alignment multiple and never receiving gradient, deliberately masked padding embeddings, an adapter matrix zero at initialization.

Each of those is a declaration. So an exact zero in a trained tensor is a receipt, and the one value in the table whose provenance is unmixed.

But safetensors is a dense container. It must write sixteen bits at every offset, so *nothing was placed here* has no way to say itself except by spending a value, and the bit pattern drafted for the job is the zero. The zero is therefore a container fact wearing the clothes of a training fact — rework 18 relocated into the file format: container, distribution, training; three layers, and the tensor reports two.

Lifting every cell until no zero remains is not a compression trick. It is making the file agree with §1.1. Zero stops being a codepoint meaning absence and returns to being the unoccupied address. Three things follow: the stored unit goes fully uniform, with no reserved pattern and no per-element test anywhere; arithmetic loses its absorbing element, so every cell's contribution remains recoverable from a result and provenance stops being mixed at the level of the operation rather than merely at the level of the code; and absence is carried structurally, by shape and offset and extent, which is the obligation the human's register already assumes.

The second of those strengthens §1.2 item 3 beyond what v0.1 claimed, and it is the reason the walk was worth six turns.

9.5 What the six turns show

§8.6 predicted that the eighteen non-camouflage reworks are catchable by intention. The second run is a test of that prediction under the most favourable conditions available: an instance holding the full document, able to cite the mechanism by name, on a substrate where the human had declared the operand set in advance.

It cost six turns anyway, and the mechanisms were not new ones. Entry 23 is the plainest: the model held the premise, stated it in the same answer, and filed it under the opposite heading. Nothing was missing. The reading was wrong because the question had been received as *what is this value* rather than *is this an address*.

The conclusion the v0.3 reading supports is narrower and worse than 8.6's. Holding the document does not remove the prior. It changes what the prior costs: the mechanisms become nameable after the fact, one turn sooner, which is the difference between nineteen reworks over twenty-four hours and six over one exchange. That is a real improvement and it is not the improvement vigilance was supposed to deliver.

What actually removed the prior, in the end, was not an argument. It was the human continuing to remove places where arithmetic could stand until there were none left — and the model, at each step, being able to see only the step it had just been moved off.

9.6 Errata to v0.2

1. **§8.6.** *The other eighteen reworks are catchable by intention* is too strong. Entries 20–24 were committed by an instance holding the list, one of them in the turn immediately after naming the risk. Amended: they are *nameable* by intention, after the fact. Catching them still requires the other party.
2. **§7.** *The record above is the only instrument available* no longer holds. The cross-instance comparison of 9.2 is a second instrument. It is also weak, and in a different way: it confounds memory, documents and conversational history unless the arms are declared in advance, which they were not.
3. Entries 1 and 2 remain unresolved and unverified at v0.3.

9.7 The declared operand set

Standing as of v0.3. Changes from 8.8 are marked; unmarked clauses are unchanged.

Admitted. Integers as the only operands. Integer division; modulus; mask; shift. Sign hoisted to plane or to lookup. One pass where one pass is specified. Lookup preferred to runtime decision throughout; a computation is evidence that a lookup was missed.

Excluded. Products. Floats and dot products. Transcendentals. Random draws. Asymptotic tests. Ordering and comparison as primitives. Recursion and termination tests where a single division was specified. **[new]** Zero as a stored value: absence is an unoccupied address, carried by geometry, and never by a codepoint in the value domain.

Flagged, not filled. The four gaps of 8.8 stand. Two are added.

1. Addition and subtraction: not yet admitted or excluded. Deferred by declaration.
2. The admissibility condition $|ad-bc| = 1$ contains two products and a subtraction, and is therefore a lookup rather than an evaluation. The form of the lookup — resident table, mask, or address test — is not yet fixed.
3. Shift is admitted and resolves into three distinct kinds, one of which is a shift of position in set space rather than a bit move. Not yet specified.
4. Equality: whether $= 1$ is an admitted test or whether satisfaction of the admissibility condition is a resident address property. Fixed once the field is normalized, not before.
5. **[new]** The lift: what the consumer does on arrival at a lifted cell. If the cell is inert, a marker skipped downstream, the exclusion above is sufficient. If it must participate in arithmetic, it needs a value and not only an address, and that is a separate declaration. Not made.
6. **[new]** Whether the four placement populations — pruned, padded, masked, never-trained — are one kind of absence or four. They are distinguishable at lift time and the distinction is discarded by a uniform exclusion. Recorded rather than resolved.